

Agentic AI in Practice

From Spec to Productive Workflow

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GöAID Session · Academic Cloud

Where I come from

- openEduSuite: sovereign groupware suite
- Bare-metal Kubernetes, in production
- AI in the suite, agents on the operation
- Real mailboxes, real students, real outages

A story from production, not a demo.

Roadmap — 30 minutes

Block	Content	Time
1 · Foundations	Model + harness + loop	~8 min
2 · Method	Spec → Contract → Test	~10 min
3 · Practice	Four real cases	~12 min

Then: 10 minutes of questions.

Agentic ≠ chat

- Chat: question → answer
- Agent: goal → plan → tools → verify
- The difference: the harness, not the model

| *"Agentic" is a working environment.*

The harness decides

- Terminal agent: files, git, cluster
- pi: MCP tools, skills, memory
- Plan mode: consent before changes

Permissions live in the contract, not in gut feeling.

pi in 60 seconds

```
curl -fsSL https://pi.dev/install.sh | sh  
npm install -g --ignore-scripts @earendil-works/pi-coding-agent
```

- v1.0: fullscreen · lean codemode · images
- **Ctrl+L** — switch model mid-session
- **Ctrl+P** favorites · **Shift+Tab** thinking
- Provider-agnostic · `/hotkeys` lists all

"Can it leak my keys?" — pi-sandbox

- Escape probes + real exfiltration test
- Recommended Docker: stops accidents only
- Hardened flags: pi won't even start
- Works: seccomp, gVisor, secrets-broker

Sandboxing is a measurement, not a checkbox.

The pyramid: Spec · Contract · Test

Level	Says	Domain cutover	SSO test suite
Spec	WHAT + acceptance	Swap domain, guards 157 + 8	15 clients e2e
Contract	HOW the agent works	Plan-mode first, idempotent	Every warning: a decision
Test	WHEN it's right	Re-run = no diff	59 checks green → merge

Every level is text in the repo.

From spec to change: OpenSpec

- Change = proposal + spec delta + tasks
- Agent implements, human reviews, archives
- Archive: main spec is the new state

```
Spec → Contract → Agent  
    → Verify (tests) → Operate (ArgoCD)
```

Documentation is the product.

OpenSpec × pi

- `pi install npm:openspec-pi`
- Agent queries spec state mid-task
- Auto-context: injected at session start
- `/opsx-new` · `apply` · `verify` · `archive`

The spec is the agent's operating context.

The stage: openEduSuite

Component	Service	Component	Service
Identity	Keycloak (SSO)	Chat	Matrix/Synapse
Groupware	SOGGo	Projects	OpenProject
Files	OpenCloud	Video	Jitsi/Intercom
Wiki	XWiki	Portal	collab-dashboard

- ArgoCD GitOps · bare metal · Galera
- Agents on production: spec first, tests accept

Case 1 · Domain migration, as a spec

- Cutover: `home.openedu` → `suite.graphwiz.ai`
- Ingresses, certificates, redirects, mail
- Contract: idempotent script, guards 157 + 8
- Agent implements · checks green · then run

Nerve-racking cutover becomes routine: just rerun.

Case 2 · SSO mystery

- Symptom: white page after SAML login
- Plan mode: hypotheses → probes → fix
- Causes: dead domains, TLS hop, ACS URL
- Stalwart: dead OIDC issuer in directory

Debugging: symptom in, root-cause chain out.

Case 3 · "Fix-all": drift to GitOps

- k8s directories without an ArgoCD app
- Legacy from imperative applies
- Orders: every app GitOps, diffs explained
- Side finding: `k8s/llm` — same suite

Boring, important work — perfect for agents.

Case 4 · Fleet with acceptance gates

- SSO suite: 59 checks, 15 clients e2e
- Built in batches by parallel workers
- Isolated worktrees, exact criteria
- Merge only on green — else discarded

Many small orders — not one super-agent.

The tool: agentflow (af)

- Rust orchestrator: `tasks.json` + `workers.json`
- Parallel workers, isolated git worktrees
- Gate: exit 0, then merge
- Retries: fresh branch + error history
- UCB1 routing, receipts in `af cost`

Contracts once — the runner enforces the gates.

What the operation learned

- Tight specs beat big contexts
- Agents need contracts, not faith
- Every incident becomes a test

Trust is good. Verification is cheaper.

Apply it to your project

1. Contract first: `AGENTS.md`
2. Spec change before code
3. CI green = done
4. Memory in the repo, not the chat

Start with one boring, important task.

Resources

- openEduSuite · openedusuite.graphwiz.ai
- pi · pi.dev — docs: pi.dev/docs
- pi-openspec · [npm:openspec-pi](https://npm.com/openspec-pi)
- pi-sandbox · evidence + threat model
- agentflow · worktrees + gates
- Workshop repo · [IT4Science-Days-2026-agent-ai-workshop](https://github.com/IT4Science-Days-2026/agent-ai-workshop)

Thank you!

Questions? — 10 minutes

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